MOLES AND GASES 2

One mole of any (ideal) gas occupies the same volume as any other (ideal) gas that is at the same temperature and pressure. **Ideal gases** are those that behave in a certain, predictable manner. Often a standard temperature and pressure (STP) or standard laboratory conditions (SLC) are quoted in reference to gases. Common standards include 1 atmosphere pressure (sometimes quoted as 101.3 kPa) and either 0°C (273 K) or 25°C (298 K).

For questions on this page assume that 1 mole of a gas at STP occupies 22.40 litres. Different sources will use different values so check carefully the values you are given to work with.

Quantities of gases can be quoted using moles, volume or mass. It is essential to be able to convert between these measurements.

$$\text{number of moles} = \frac{\text{mass present}}{\text{molar mass}} = \frac{\text{volume present}}{\text{molar volume}}$$

EXAMPLES

i **How many moles of gas are present in 12.00 g of carbon dioxide?**

Carbon dioxide has the formula CO_2: one carbon atom (12×12.01) and two oxygen atoms (2×16.00)

Formula mass: $(12.01 + 32.00) = 44.01$ amu
Hence one mole of carbon dioxide has a mass of 40.02 g.

The number of moles is calculated by dividing the mass present (12.00 g) by the molar mass (44.01 g) = 0.2727 moles

ii **How many moles of sulfur dioxide are in 500 mL of the gas at STP?**

Assume that one mole of sulfur dioxide occupies 22.40 litres. The number of moles is calculated by dividing the volume present (500 mL = 0.5 L) by the molar volume (22.40 L) = 0.022 moles.

1 Calculate the moles of gas present in 15.50 g of ammonia (NH_3).

2 Calculate the mass of 0.25 moles of oxygen molecules.

3 Calculate the number of moles (at STP) of 1.76 litres of sulfur trioxide.

4 Calculate the mass of 12.00 litres of hydrogen molecules. (Hint: you need to do two calculations.)

5 What would be the volume of 3.00 kg of butane (C_4H_{10}) at STP?

At constant temperature and pressure, the volume of a gas is proportional to the number of moles; therefore chemical equations involving gases deal with volumes of gas as well as numbers of moles.

Look at the equation for the manufacture of ammonia from nitrogen and hydrogen:

$N_2\,(g) + 3H_2\,(g) \rightarrow 2NH_3\,(g)$

This equation can be read in several ways:

- One molecule of nitrogen reacts with three molecules of hydrogen to form two molecules of ammonia
- One mole of nitrogen reacts with three moles of hydrogen to form two moles of ammonia
- One volume of nitrogen reacts with three volumes of hydrogen to form two volumes of ammonia
- One litre of nitrogen will react with three litres of hydrogen to form two litres of ammonia.

6 Methane, CH_4, ('natural gas') is used as a fuel in household gas heaters and stoves. Methane burns in air to form carbon dioxide and water.

a Write a balanced equation to show the combustion of methane.

b How many moles of oxygen are needed to burn one mole of methane?

c How many litres of carbon dioxide are formed when 100 litres of methane is burned?

7 Sulfur dioxide, SO_2, can react with oxygen to form sulfur trioxide, SO_3.

a Write a balanced equation for the reaction between sulfur dioxide and oxygen.

b How many moles of oxygen are needed per mole of sulfur dioxide?

c How many litres of oxygen are needed to react with 50 litres of sulfur dioxide?

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